Wei Wang

Ph.D. Candidate in Mathematics, School of Mathematical Sciences, Peking University

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Education

Sept. 2022 – 2027 (expected)
- **Peking University, School of Mathematical Sciences**
  Ph.D. Candidate in Mathematics
  Beijing, China
  Advisor: Prof. Zhifei Zhang

Sept. 2018 – June 2022
- **Zhejiang University, School of Mathematical Sciences**
  B.S. in Mathematics and Applied Mathematics
  Hangzhou, Zhejiang, China
  Advisor: Prof. Ting Zhang
  Thesis: *Homogenization Problem of Second Order Elliptic Systems with Lower Order Terms*

Research Interests

Geometric variational problems; elliptic and parabolic partial differential equations; harmonic maps and their flows; liquid crystal theory; homogenization theory; nonlocal operators.

Publications and Accepted Papers

1. Sharp convergence rates for parabolic Green's functions in time-independent periodic homogenization.
   Wei Wang. Accepted for publication in *Frontiers of Mathematics*. [arXiv: 2606.12101]
2. Landau–de Gennes model with sextic potentials: asymptotic behavior of minimizers.
   Wei Wang and Zhifei Zhang. Accepted for publication in *Memoirs of the European Mathematical Society*. [arXiv: 2404.00677]
3. Existence of weak solutions for two-phase matrix-valued harmonic map flows.
   Wei Wang, Wei Wang (王伟), and Zhifei Zhang. *Journal of Functional Analysis* 291 (2026), no. 11, Article No. 111623. [doi] [arXiv: 2506.16278]
4. Uniform estimates of Landau–de Gennes minimizers in the vanishing elasticity limit with line defects.
   Haotong Fu, Huaijie Wang, and **Wei Wang**. *SIAM Journal on Mathematical Analysis* 58 (2026), no. 4, 3306–3324. [doi] [arXiv: 2508.01811]
5. Stratification and rectifiability of harmonic map flows via tangent measures.
   Haotong Fu, **Wei Wang**, Ke Wu, and Zhifei Zhang. *Mathematische Annalen* 395 (2026), Article No. 92. [doi] [arXiv: 2504.14880]
6. On the sharp constants in the regional fractional Sobolev inequalities.
   Rupert L. Frank, Tianling Jin, and **Wei Wang**. *SN Partial Differential Equations and Applications* 6 (2025), Article No. 15. [doi] [arXiv: 2403.00357]
7. Uniform estimates of resolvents in homogenization theory of elliptic systems.
   Wei Wang. *Journal of Differential Equations* 370 (2023), 1–65. [doi] [arXiv: 2211.14614]
8. Homogenization theory of elliptic systems with lower order terms in dimension two.
   Wei Wang and Ting Zhang. *Communications on Pure and Applied Analysis* 22 (2023), no. 3, 787–824. [doi] [arXiv: 2210.06329]

Preprints and Manuscripts

1. Finiteness of singularities for higher-weight axially symmetric minimizers of the relaxed Dirichlet energy.
   Zhiyuan Dai, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2611.07414]
2. Uniform boundary observability for periodic waves, I: up to the ε^{-1} eigenvalue scale.
   Jun Geng, Shumao Wang, and **Wei Wang**. Preprint. [arXiv: 2610.33547]

3. Asymptotics for two-dimensional parabolic vectorial Allen–Cahn systems.
   Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2609.27596]
4. Small-energy regularity for stationary weak solutions of the Liouville equation.
   Haotong Fu, Kelei Wang, **Wei Wang**, and Ke Wu. Preprint. [arXiv: 2609.26026]
5. Boundary asymptotics for two-dimensional vectorial Allen–Cahn systems.
   Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2609.00739]
6. Boundary regularity for Oseen–Frank minimizers with arbitrary positive splay, twist, and bend constants.
   Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2608.22287]
7. Sharp instability of planar isotropic–nematic interfaces in the Landau–de Gennes model.
   Wei Wang and Qin Wu. Preprint. [arXiv: 2608.12131]
8. Smooth convergence of Landau–de Gennes minimizers in the Lyuksyutov regime.
   Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2608.08456]
9. On the absence of point defects in biaxial Landau–de Gennes models.
   Haotong Fu, Huaijie Wang, **Wei Wang**, and Zhifei Zhang. Preprint. [arXiv: 2608.04908]
10. Singular limits for three-dimensional global minimizers of Ginzburg–Landau-type functionals: uniform estimates and singular sets.
    Giacomo Canevari, Haotong Fu, and **Wei Wang**. Preprint. [arXiv: 2606.27691]
11. A note on an approximate fixed point theorem for VMO maps.
    Haotong Fu and **Wei Wang**. Preprint. [arXiv: 2606.15670]
12. Asymptotics of minimizers for Ginzburg–Landau-type functionals in high dimensions.
    Giacomo Canevari, Haotong Fu, and **Wei Wang**. Preprint. [arXiv: 2605.04442]
13. Improved convergence of Landau–de Gennes minimizers in the vanishing elasticity limit.
    Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2507.14955]
14. Fine structure of rupture set for semilinear elliptic equations with singular nonlinearity.
    Wei Wang and Zhifei Zhang. Preprint. [arXiv: 2411.16048]
15. Quantitative stratification and sharp regularity estimates for supercritical semilinear elliptic equations.
    Haotong Fu, **Wei Wang**, and Zhifei Zhang. Preprint. [arXiv: 2408.06726]

AI for Math Projects

1. **Paper Math Bench: research-based benchmark construction**
   Designed a reusable AI skill linking paper analysis, difficult calculation problems, step-by-step reference solutions, independent review, and model evaluation. Specified task interfaces and mathematical checks; retained version-matched questions, solver outputs, grading, and usage records. [GitHub] [Blog]
2. **Minimizing harmonic maps: regularity and stratification in Lean**
   Directed AI-assisted formalization of small-energy regularity and singular-set stratification. Supplied mathematical proofs, decomposed dependencies, and reviewed theorem interfaces; extracted explicit recurrences for regularity constants. [Regularity] [Stratification] [Blog]
3. **Stationary harmonic maps: regularity and explicit constants**
   Formalized small-energy regularity in Lean via the Rivière–Struwe gauge method for general compact targets. Computed exact rational bounds for the energy threshold and Hölder estimate, with explicit dependence on geometric inputs. [GitHub] [Blog]

Research Visits and Programs

Mar. – Apr. 2026 • **University of Verona, Department of Computer Science**
Visiting Ph.D. Student
Verona, Italy
Host: Prof. Giacomo Canevari

June – Sept. 2021 • **HKUST, Department of Mathematics**
Summer Research Program (Online)
Hong Kong, China
Advisor: Prof. Tianling Jin

Selected Talks

Aug. 2026 • Conference on Geometric Measure Theory and Partial Differential Equations
Enshi, Hubei, China.

July 2026 • Academic Forum on Theory and Numerical Analysis of Liquid Crystals and Complex Systems
Xiangtan University.

Jan. 2026 • Academic Workshop on Fluid Dynamics Equations
Peking University.

Nov. 2025 • Jinhua Conference on Harmonic Analysis and Its Applications
Zhejiang Normal University.

Nov. 2025 • 3rd National Academic Exchange Forum for Doctoral Students in Frontier Mathematics
Peking University, Beijing, China.

Aug. 2025 • Conference on Theory and Computation of Quasi Crystals
Nankai University.

June 2025 • PDE and Geometry Seminar
Zhejiang University.

Dec. 2024 • Workshop of the PDE Group
Peking University, Beijing, China.

Grants and Projects

2023 – 2028 • **Physical Modelling and Mathematical Theory of Liquid Crystals**
Participant, NSFC, Grant No. 12288101.

2023 – 2028 • **Physical Modeling and Mathematical Theory of Liquid Crystal Systems**
Participant, National Key R&D Program of China, Grant No. 2023YFA1008801.

Academic Service

- **Mathematical Reviews / MathSciNet, American Mathematical Society**
  Registered prospective reviewer.
- **Academic Committee, 4th Frontiers Doctoral Forum**
  Screening and organization of talks in Analysis and Differential Equations.

Teaching Experience

2022 – 2026 • **Peking University, Teaching Assistant**
Advanced Mathematics (A, B, and C); Real Analysis; Functional Analysis (I); Harmonic Analysis.

2021 – 2022 • **Zhejiang University, Teaching Assistant**
Mathematical Analysis (I and II).

Honors and Awards

- Sept. 2026 • **National Scholarship**, School of Mathematical Sciences, Peking University.
- Apr. 2025 • **First Prize, Graduate Research Exhibition**, School of Mathematical Sciences, Peking University.
- Apr. 2024 • **Third Prize, Graduate Research Exhibition**, School of Mathematical Sciences, Peking University.
- Oct. 2023 • **Elite Program for Graduate Students in Mathematics**, Peking University.
- Aug. 2022 • **Nomination Award, ICCM Creative Undergraduate Thesis**, Shing-Tung Yau Center of Southeast University.
- Apr. 2021 • **First Prize, 11th Chinese Mathematics Competitions Finals**, Wuhan University.
- June 2021 • **Nomination Award, S.-T. Yau College Student Mathematics Contest**, Yau Mathematical Sciences Center, Tsinghua University.

Skills and Languages

- Skills • LaTeX, MATLAB, Mathematica, Python, Lean.
- Languages • Chinese (native), English (fluent).